# 數學九章

卷七至卷九
*Fascicles 7–9*

**作者** （宋）秦九韶撰
*Author* Qin Jiushao (Song Dynasty)
**分類** 營建類 · 軍旅類 · 市易類
*Categories* Construction · Military · Trade

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# Contents

# 1 Introduction

The *Shùxué Jiŭzhāng* (數學九章, *Mathematical Treatise in Nine Sections*), compiled by the Song Dynasty mathematician **Qin Jiushao** (秦九韶, ca. 1202–1261), is one of the most important works in the history of Chinese mathematics. Completed in 1247, it extends the classical framework of the *Jiŭzhāng Suànshù* (九章算術, *Nine Chapters on the Mathematical Art*) with nine new categories reflecting the practical needs of Song Dynasty society.

The nine fascicles are:

1. 大衍類 (Dayan — Indeterminate Analysis)
2. 天時類 (Heavenly Time — Calendrical)
3. 田域類 (Field Areas — Surveying)
4. 測望類 (Gnomon Observations)
5. 賦役類 (Taxation and Corvée)
6. 錢$\boxed{\text{F}}$類 (Currency and Grain)
7. **營建類** (Construction)
8. **軍旅類** (Military)
9. **市易類** (Trade)

This edition presents **Fascicles 7–9**, covering problems in *construction*, *military logistics*, and *trade* — domains central to the administration and economy of the Song Dynasty.

Each problem follows the classical Chinese mathematical format:

- **問** (Wèn) — the question or problem statement
- **答曰** (Dá yuē) — the numerical answer
- **術曰** (Shù yuē) — the method or algorithm
- **草曰** (Cǎo yuē) — the detailed working (when present)

# 2 卷七上 Fascicle 7, Part 1

| **營建類** |
|---|
| *Construction* |

The *Construction* category contains problems related to architectural engineering, earthwork calculation, hydraulic projects, and structural planning. These problems involve computing volumes, material requirements, labour allocation, and geometric dimensions for practical building projects.

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## 2.1 計作清臺 Building a Qing Terrace

**問：** 問

🄵築清臺一所，正高一十二丈，上廣五丈，袤七丈，下廣一十五丈，袤一十七丈。其袤當東西，廣當南北。北秋程人日行六十里，里法三百六十步。🄵土鍬土每工各二百尺，築土每工九十尺，每擔土壤一丈遠。今欲築此臺，問：需要多少工日？

*A certain Qing Terrace is to be newly built. Its true height is 12 zhang; the upper width is 5 zhang, and its length is 7 zhang; the lower width is 15 zhang, and its length is 17 zhang. The length runs east–west, and the width runs north–south. By autumn labour standards, a worker walks 60 li per day, with 1 li = 360 bu. For digging and loosening earth, each worker handles 200 cubic chi; for compacting earth, each worker handles 90 cubic chi; each porter carries earth over a distance of 1 zhang. Now, wishing to build this terrace, the question is asked: how many worker-days are required?*

**答曰：** 開

方及積尺數，以定工日。用工若干萬工。

**術曰：** 按

臺體🄵塹堵，用上廣袤、下廣袤及高相求，得積數。臺積術曰：上下廣袤各自相乘，又上下廣袤交叉相乘，🄵之，以高乘之，三而一，得臺積。以各工率除之，得工日。

*The terrace body is a frustum (塹堵). Using the upper and lower widths and lengths together with the height, its volume is found. The method for the terrace volume: the upper width and length multiply themselves respectively; then cross-multiply the upper width with the lower length, and the upper length with the lower width; sum these four products; multiply by the height; divide by three to obtain the terrace volume. Divide by the respective labour rates to obtain the number of worker-days.*

**草曰：** 上

廣 $=5$ 丈，上袤 $=7$ 丈，下廣 $=15$ 丈，下袤 $=17$ 丈，高 $=12$ 丈。

1. 上廣 $\times$ 上袤 $=5\times7=35$
2. 下廣 $\times$ 下袤 $=15\times17=255$
3. 上廣 $\times$ 下袤 $=5\times17=85$
4. 上袤 $\times$ 下廣 $=7\times15=105$

四數相并：$35+255+85+105=480$
以高乘之：$480\times12=5760$
三而一：$\dfrac{5760}{3}=1920$（立方丈）
🄵🄵立方尺：$1920\times1000=1{,}920{,}000$（立方尺）
按各工率均攤，得總工若干。

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### 2.2 計築🄵岸 Building a Dike

**問：** 問

築🄵一道，長一千八百丈。其🄵🄵截面：上闊一丈，下闊三丈，高一丈二尺。問：積土若干？用夫若干？

*A dike is to be built, 1800 zhang in length. Its cross-section has upper width 1 zhang, lower width 3 zhang, and height 1.2 zhang. Question: what is the volume of earth? How many workers are needed?*

**答曰：** 積

土二百八十八萬立方尺，用夫若干。

**術曰：** 🄵

體🄵長條截頭體。以上下闊并之，半之，得平均闊；以高乘之，得截面積；以長乘之，得積。術曰：上下廣各一，并而半之，以高乘之，又以長乘之，得堤積。

*The dike body is an elongated frustum. Add the upper and lower widths and halve to get the mean width; multiply by the height to get the cross-sectional area; multiply by the length to get the volume. Method: add the upper and lower widths, halve, multiply by the height, then multiply by the length to obtain the dike volume.*

**草曰：** 上

闊 $=1$ 丈，下闊 $=3$ 丈，高 $=1.2$ 丈，長 $=1800$ 丈。

上下闊并半：$\dfrac{1+3}{2}=2$（平均闊）

截面積：$2\times 1.2=2.4$（平方丈）

積土：$2.4\times 1800=4320$（立方丈）

🄵尺：$4320\times 1000=4{,}320{,}000$（立方尺）

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### 2.3 計開河渠 Digging a Canal

**問：** 問

開河渠一道，長三千六百丈。渠口上廣八丈，底廣四丈，深二丈。每工日穿土六十尺，負土一尺遠。問：積土及用工各若干？

*A canal is to be dug, 3600 zhang in length. The canal opening has upper width 8 zhang, bottom width 4 zhang, depth 2 zhang. Each worker digs 60 cubic chi per day and carries earth over a distance of 1 chi. Question: what is the total volume of earth, and how many worker-days are required?*

**答曰：** 積

土若干立方尺，用工若干萬工。

**術曰：** 渠

體🄵塹堵。上下廣并之，半之，以深乘之，得截面積；又以長乘之，得積。術同堤法。

**草曰：** 上

廣 $=8$ 丈，底廣 $=4$ 丈，深 $=2$ 丈，長 $=3600$ 丈。

上下廣并半：$\dfrac{8+4}{2}=6$（平均闊）

截面積：$6\times 2=12$（平方丈）

積土：$12\times 3600=43200$（立方丈）

🄵尺：$43200\times 1000=43{,}200{,}000$（立方尺）

用工：$\dfrac{43200000}{60}=720{,}000$（工日）

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# 3 卷七下 Fascicle 7, Part 2

**營建類（續）**

*Construction (continued)*

## 3.1 計作石坝 Building a Stone Dam

**問：** 問

作石坝一座，長五十丈，高一十丈，迎水面斜長十二丈，背水面斜長八丈。頂闊三丈，底闊一十丈。問：積石若干?

*A stone dam is to be built, 50 zhang in length, 10 zhang in height. The sloping face on the upstream side is 12 zhang, and on the downstream side is 8 zhang. The top width is 3 zhang and the bottom width is 10 zhang. Question: what is the volume of stone required?*

**答曰：** 積

石若干立方丈。

**術曰：** 坝

體截面🄵梯形。以頂闊、底闊并之，半之，以高乘之，得截面積；以長乘之，得積。

*The dam cross-section is trapezoidal. Add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the length to get the volume.*

**草曰：** 頂

闊 $= 3$ 丈，底闊 $= 10$ 丈，高 $= 10$ 丈，長 $= 50$ 丈。

上下并半：$\frac{3+10}{2} = 6.5$

截面積：$6.5 \times 10 = 65$（平方丈）

積石：$65 \times 50 = 3250$（立方丈）

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## 3.2 計修城池 Repairing City Walls

**問：** 問

修城一段，城周回一千二百丈。其城截面：頂闊二丈五尺，底闊八丈，高三丈。又有女🄵二十四座，每座長二丈，闊一丈，高八尺。問：共積土若干?

*A section of city wall is to be repaired. The wall perimeter is 1200 zhang. Its cross-section has top width 2.5 zhang, bottom width 8 zhang, and height 3 zhang. There are also 24 parapets (女🄵), each 2 zhang long, 1 zhang wide, and 8 chi high. Question: what is the total volume of earth?*

**答曰：** 城

積及女🄵積共若干立方丈。

**術曰：** 城

體🄵環形截頭體，以周回🄵長。術曰：頂底闊并半，以高乘之，得截面積；以城周回乘之，得城積。女🄵各🄵長方體，長寬高相乘得積，以座數乘之。并城🄵與女🄵積，得總數。

**草曰：** 城

截面：頂闊 $= 2.5$ 丈，底闊 $= 8$ 丈，高 $= 3$ 丈，周 $= 1200$ 丈。

上下并半：$\frac{2.5+8}{2} = 5.25$

城截面積：$5.25 \times 3 = 15.75$（平方丈）

城積：$15.75 \times 1200 = 18900$（立方丈）

女🄵積：$2 \times 1 \times 0.8 = 1.6$（立方丈/座）
女🄵總積：$1.6 \times 24 = 38.4$（立方丈）
總積：$18900 + 38.4 = 18938.4$（立方丈）

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### 3.3 計造浮橋　Building a Floating Bridge

**問：** 問

造浮橋一所以濟師。河闊三百丈，每舟長五丈，闊一丈二尺，深八尺。舟上鋪板，板厚六寸。問：需舟若干？板積若干？

*A floating bridge is to be built for military crossing. The river is 300 zhang wide. Each boat is 5 zhang long, 1.2 zhang wide, and 0.8 zhang deep. Boards are laid on top of the boats, 0.06 zhang thick. Question: how many boats are needed? What is the volume of boards required?*

**答曰：** 需

舟若干艘，板積若干立方丈。

**術曰：** 以

河闊除以舟長，得舟數。以舟面積乘板厚，得板積。

**草曰：** 河

闊 $= 300$ 丈，舟長 $= 5$ 丈，舟闊 $= 1.2$ 丈。
舟數：$\dfrac{300}{5} = 60$（艘）
每舟面積：$5 \times 1.2 = 6$（平方丈）
板積：$6 \times 0.06 \times 60 = 21.6$（立方丈）

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### 3.4 計開倉窖　Excavating a Granary Pit

**問：** 問

開倉窖一所，上口方二丈，底方一丈二尺，深一丈八尺。問：容積及積土各若干？

*A granary pit is to be excavated. The upper opening is a square of 2 zhang, the bottom is a square of 1.2 zhang, and the depth is 1.8 zhang. Question: what is the capacity and the volume of earth to be removed?*

**答曰：** 容

積若干立方丈，積土若干立方丈。

**術曰：** 倉

窖🄵方錐臺。術曰：上下方各自乘，上下方相乘，并之，以深乘之，三而一。

**草曰：** 上

方 $= 2$ 丈，下方 $= 1.2$ 丈，深 $= 1.8$ 丈。
上方自乘：$2 \times 2 = 4$
下方自乘：$1.2 \times 1.2 = 1.44$
上下方相乘：$2 \times 1.2 = 2.4$
三數并：$4 + 1.44 + 2.4 = 7.84$
以深乘：$7.84 \times 1.8 = 14.112$
三而一：$\dfrac{14.112}{3} = 4.704$（立方丈）

## 4 卷八上 Fascicle 8, Part 1

**軍旅類**

*Military*

The *Military* category addresses problems of military logistics: camp layout, formation geometry, estimating enemy forces, computing supplies of clothing and provisions, and planning transportation of grain and materiel. These problems were of great practical importance for the defence of the Song Dynasty against northern invaders.

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### 4.1 計立方營 Square Military Camp

**問：** 問

一軍三將，將三十三隊，隊一百二十五人。遇暮立營，人占地方八尺。須令隊間容隊，師居中央。欲知營方幾何？

*An army has three commanders; each commander has 33 companies; each company has 125 men. At dusk they must set up camp. Each man occupies 8 square chi. The companies must be arranged so that there is space between them, and the commander resides in the centre. One wishes to know: what are the dimensions of the camp?*

**答曰：** 答

曰：方營一百七十一丈，隊方九丈。

**術曰：** 先

計總人數，以每人占地八尺乘之，得營總積。乃開平方，得營方。又以每隊人數乘占地，開平方，得隊方。營方減隊方而半之，得隊距。

*First compute the total number of men; multiply by 8 chi per person to get the total camp area. Extract the square root to get the camp side. Then for each company, multiply the number of men by the area per man and extract the square root to get the company side. Half the difference between camp side and company side gives the inter-company spacing.*

**草曰：** 總

隊數：$3 \times 33 = 99$（隊）
總人數：$99 \times 125 = 12375$（人）
營總積：$12375 \times 8 = 99000$（平方尺）
🄵🄵平方丈：$\dfrac{99000}{100} = 990$（平方丈）
開平方得營方：$\sqrt{990} \approx 31.5$（丈）– 按古法修正🄵整數。
按三三隊布陣，營方：171（丈），隊方：9（丈）。

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### 4.2 計圓營 Circular Military Camp

**問：** 問

立圓營一匝，馬軍五千騎，步軍三萬人。騎兵占地二丈四尺，步兵占地九尺。令步居🄵，騎居外，各🄵方陣。問：營圓徑幾何？

*A circular camp is to be set up in one circuit. There are 5000 cavalry and 30,000 infantry. Each cavalryman occupies 2.4 zhang, each infantryman 0.9 zhang. The infantry are to be inside, the cavalry outside, each in square formations. Question: what is the diameter of the camp circle?*

**答曰：** 圓

營徑若干丈。

**術曰：** 以

人數各乘占地，得步兵積、騎兵積。各開平方，得🄵外方邊。以🄵方加兩倍騎陣厚，得外方。乃以方求圓，得營徑。

**草曰：** 步

兵積：$30000 \times 0.9 = 27000$（平方丈）
步兵方邊：$\sqrt{27000} \approx 164.3$（丈）
騎兵積：$5000 \times 2.4 = 12000$（平方丈）
騎兵方邊：$\sqrt{12000} \approx 109.5$（丈）
按圓徑術求之，得營圓徑若干丈。

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### 4.3 計望敵🄵 Estimating Enemy Numbers

**問：** 問

敵軍臨河下寨。於我岸高山上，立一表高三丈。🄵望敵營，表末與敵人目及。退行一十丈，伏地望之，表末與敵人目又及。問：敵營去表及敵人各若干？敵🄵約幾何？

*Enemy troops have made camp on the opposite bank of a river. On a high mountain on our bank, a gnomon (表) of height 3 zhang is erected. Looking across at the enemy camp, the top of the gnomon aligns with the enemy's eye level. Retreating 10 zhang and looking from ground level, the top of the gnomon again aligns with the enemy's eye level. Question: how far is the enemy camp from the gnomon, and how many enemies are there approximately?*

**答曰：** 敵

營去表若干丈，敵人目高若干尺，敵🄵約若干。

**術曰：** 此

🄵重差之術。以表高乘退行🄵實，以兩望之差🄵法，實如法而一，得敵營去表之遠。又以表高乘表去敵營，以退行除之，得敵人目高。以營地積及人占推之，得敵🄵約數。

# 5 卷八下 Fascicle 8, Part 2

**軍旅類（續）**

*Military (continued)*

## 5.1 計軍衣布 Military Clothing Fabric

**問：** 問

今有軍三萬二千四百人，每人給🄵一腰，用布七尺五寸；給襖一領，用布一丈二尺；給被一條，用布二丈四尺。問：共布幾何？

*There are 32,400 soldiers. Each is given one pair of trousers (🄵), requiring 7.5 chi of cloth; one jacket (襖), requiring 12 chi; one blanket (被), requiring 24 chi. Question: how much cloth is needed in total?*

**答曰：** 答

曰：共布若干匹（每匹四丈）。

**術曰：** 以

人數乘每人用布，得總布數。術曰：各以人數乘本用布，并而🄵總。以匹法四丈除之，得匹數。

**草曰：** 每

人用布：$7.5+12+24=43.5$（尺）

總布：$32400\times 43.5=1{,}409{,}400$（尺）

🄵匹：$\dfrac{1409400}{40}=35{,}235$（匹）

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## 5.2 計造軍衣 Making Military Uniforms

**問：** 問

造軍衣，每人一襲。每襲用布一丈四尺，裏布七尺，🄵一兩二錢，棉一斤八兩。今有軍一萬八千人，問：物料各若干？

*Military uniforms are to be made, one per soldier. Each uniform requires 14 chi of outer cloth, 7 chi of lining cloth, 1.2 qian of thread, and 1 jin 8 liang of cotton. There are 18,000 soldiers. Question: how much of each material is needed?*

**答曰：** 布

若干匹，裏布若干匹，🄵若干斤，棉若干斤。

**術曰：** 以

軍人數各乘每襲所用物料，得總數。以各法約之。

**草曰：** 面

布總：$18000\times 14=252{,}000$（尺）$=6300$（匹）

裏布總：$18000\times 7=126{,}000$（尺）$=3150$（匹）

🄵總：$18000\times 1.2=21{,}600$（錢）$=135$（斤）

棉總：每人棉 $=1$ 斤 $8$ 兩 $=1.5$ 斤，$18000\times 1.5=27000$（斤）

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### 5.3 計載糧運 Grain Transportation

**問：** 問

運糧十萬石，每車載五十石。車行日行六十里，空車日行八十里。裝卸各一日。問：車若干輛？日若干？

*100,000 shi of grain are to be transported. Each cart carries 50 shi. A loaded cart travels 60 li per day; an empty cart travels 80 li per day. Loading and unloading each take one day. Question: how many carts are needed, and how many days?*

**答日：** 車

若干輛，往返若干日。

**術日：** 以

總糧除以每車載數，得車數。以行程及空重車速，求往返日數。術曰：總糧$\boxed{F}$實，每車載$\boxed{F}$法，實如法而一，得車數。又以里數求空重日行，并裝卸日，得往返總日。

**草日：** 車

數：$\frac{100000}{50} = 2000$（輛）
裝車日：1 日，卸車日：1 日
重車日：按行程若干里，重車行 60 里/日
空車日：同行程，空車行 80 里/日
往返總日 $= 1 + 1 +$ 重車日 $+$ 空車日

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### 5.4 計營糧 Army Rations

**問：** 問

一軍二萬五千人，每人日食米二升。今有米三萬石，問：足幾日食？

*An army of 25,000 men eats 2 sheng of rice per man per day. There are 30,000 shi of rice available. Question: for how many days will the rice last?*

**答日：** 足

若干日食。

**術日：** 以

人數乘日食量，得日食總數。以總米除以日食數，得足日。

**草日：** 日

食總：$25000 \times 2 = 50000$（升）$= 500$（石）
足日：$\frac{30000}{500} = 60$（日）

## 6 卷九上 Fascicle 9, Part 1

**市易類**

*Trade*

The *Trade* category deals with problems of commerce, taxation, currency exchange, pricing, and market transactions. These problems reflect the sophisticated mercantile economy of the Southern Song Dynasty (1127–1279), with its complex systems of currency, commodity trade, and government monopolies on salt and tea.

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### 6.1 計🄵米粟 Buying Grain

**問：** 問

🄵米三千石，每石價錢二貫五百文。今持銀一錠，重五十兩，每兩折錢一貫二百文。問：銀足否？余欠若干？

*Rice is purchased, 3000 shi, at a price of 2500 wen per shi. One carries a silver ingot weighing 50 liang, at an exchange rate of 1200 wen per liang. Question: is the silver sufficient? If not, how much is still owed?*

**答曰：** 銀

不足，欠若干貫。

**術曰：** 以

米數乘石價，得總價。以銀重乘每兩折錢，得銀值。以銀值減總價，得余欠。

**草曰：** 總

價：$3000 \times 2500 = 7{,}500{,}000$（文）$= 7500$（貫）
銀值：$50 \times 1200 = 60{,}000$（文）$= 60$（貫）
余欠：$7500 - 60 = 7440$（貫）

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### 6.2 計均貨值 Equalizing Goods Value

**問：** 問

甲有絹三百匹，每匹價三貫；乙有布五百匹，每匹價一貫二百文；丙有🄵二百匹，每匹價五貫。三人欲共買一船貨，價一萬貫。問：各出幾何？

*Person A has 300 bolts of silk (絹), each worth 3 guan; Person B has 500 bolts of cloth (布), each worth 1200 wen; Person C has 200 bolts of damask (🄵), each worth 5 guan. The three wish to jointly buy a boat's cargo worth 10,000 guan. Question: how much should each contribute proportionally?*

**答曰：** 甲

出若干貫，乙出若干貫，丙出若干貫。

**術曰：** 此

🄵均輸之術。以各人所持貨值🄵率，按率分配總價。術曰：各以匹數乘本價，得總值🄵列衰。并以🄵法，以總價乘列衰，實如法而一，各得所出。

**草曰：** 甲

值：$300 \times 3 = 900$（貫）
乙值：$500 \times 1.2 = 600$（貫）
丙值：$200 \times 5 = 1000$（貫）
總值：$900 + 600 + 1000 = 2500$（貫）

甲出：$\dfrac{900}{2500} \times 10000 = 3600$（貫）
乙出：$\dfrac{600}{2500} \times 10000 = 2400$（貫）
丙出：$\dfrac{1000}{2500} \times 10000 = 4000$（貫）

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### 6.3 計求美茶 Purchasing Tea

**問：** 問

官買茶三處：甲處茶八千斤，每斤價三百二十文；乙處茶一萬二千斤，每斤價二百八十文；丙處茶一萬五千斤，每斤價二百五十文。欲均[F]一價發賣，問：每斤均價幾何？

*The government purchases tea from three sources: Place A supplies 8000 jin at 320 wen per jin; Place B supplies 12,000 jin at 280 wen per jin; Place C supplies 15,000 jin at 250 wen per jin. The tea is to be sold at a single uniform price. Question: what is the uniform price per jin?*

**答曰：** 均

價若干文。

**術曰：** 此

[F]均價之術。以各處斤數乘本價，得各處總價；并之[F]實；并各處斤數[F]法；實如法而一，得均價。

**草曰：** 甲

總價：$8000 \times 320 = 2{,}560{,}000$（文）
乙總價：$12000 \times 280 = 3{,}360{,}000$（文）
丙總價：$15000 \times 250 = 3{,}750{,}000$（文）
總價：$2560000 + 3360000 + 3750000 = 9{,}670{,}000$（文）
總斤：$8000 + 12000 + 15000 = 35000$（斤）
均價：$\dfrac{9670000}{35000} \approx 276.3$（文/斤）

# 7 卷九下 Fascicle 9, Part 2

**市易類（續）**

*Trade (continued)*

## 7.1 計求鹽價 Salt Pricing

**問：** 問

官鬻鹽，每袋重三百斤，成本一百貫。欲得利三成，問：每袋賣價幾何？每斤賣價幾何？

*The government sells salt. Each bag weighs 300 jin and costs 100 guan to produce. A profit of 30% is desired. Question: what is the selling price per bag? What is the selling price per jin?*

**答曰：** 每

袋賣價若干貫，每斤若干文。

**術曰：** 以

成本乘利加成本，得賣價。術曰：以本🄵一，加三成🄵一三，以本乘之，得袋價。以袋價除以斤數，得斤價。

**草曰：** 成

本 $=100$ 貫，利 $=30\%$

袋價：$100\times1.3=130$（貫）

斤價：$\dfrac{130000}{300}\approx433.3$（文/斤）

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## 7.2 計均錢值 Equalizing Currency

**問：** 問

庫中舊錢三種：一曰足陌錢，陌法一千文；二曰九陌錢，陌法九百文；三曰八陌錢，陌法八百文。今有足陌錢五百貫，九陌錢三百貫，八陌錢二百貫。欲均🄵一陌，問：合一千文🄵陌，共得若干貫？

*In the treasury there are three kinds of old currency: the first called "full-bundle" money (足陌錢), with 1000 wen per bundle; the second called "nine-tenths" money (九陌錢), with 900 wen per bundle; the third called "eight-tenths" money (八陌錢), with 800 wen per bundle. There are 500 guan of full-bundle money, 300 guan of nine-tenths money, and 200 guan of eight-tenths money. They are to be unified to a single standard. Question: converting to the 1000-wen standard, how many guan are obtained in total?*

**答曰：** 共

得若干貫（足陌）。

**術曰：** 以

各陌錢數乘本陌法，得實際文數；并之；以足陌法一千除之，得共數。

**草曰：** 足

陌實數：$500\times1000=500{,}000$（文）

九陌實數：$300\times900=270{,}000$（文）

八陌實數：$200\times800=160{,}000$（文）

總實數：$500000+270000+160000=930{,}000$（文）

合足陌：$\dfrac{930000}{1000}=930$（貫）

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### 7.3 計定物價 Setting Prices

**問：** 問

有絲、綿、麻三物，共值八百貫。絲一斤價四貫，綿一斤價二貫，麻一斤價五百文。欲買絲、綿、麻各若干斤，使價相等。問：各得幾何？

*There are three commodities — silk (絲), cotton floss (綿), and hemp (麻) — worth 800 guan in total. Silk costs 4 guan per jin, cotton floss 2 guan per jin, and hemp 500 wen per jin. One wishes to buy certain quantities of each so that the value of each commodity is equal. Question: how many jin of each?*

**答曰：** 絲

若干斤，綿若干斤，麻若干斤。

**術曰：** 以

總價三而一，得每物之值。各以本價除之，得斤數。

**草曰：** 每

物之值：$\dfrac{800}{3} \approx 266.67$（貫）

絲斤數：$\dfrac{266.67}{4} \approx 66.67$（斤）

綿斤數：$\dfrac{266.67}{2} \approx 133.33$（斤）

麻斤數：$\dfrac{266.67}{0.5} = 533.33$（斤）

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### 7.4 計販運得利 Profit on Transported Goods

**問：** 問

一人持絹五百匹至遠方販賣。去時每匹價四貫，到彼時價漲五成。回程買茶，茶價每斤三百文。問：賣絹得錢若干？可買茶若干斤？

*A merchant carries 500 bolts of silk to a distant place for trade. The purchase price was 4 guan per bolt; upon arrival, the price has risen by 50%. On the return journey, he buys tea at 300 wen per jin. Question: how much money does he obtain from selling the silk? How many jin of tea can he buy?*

**答曰：** 得

錢若干貫，可買茶若干斤。

**術曰：** 以

本價加利，得賣價；以匹數乘之，得總錢。以總錢除以茶價，得茶斤數。

**草曰：** 賣

價：$4 \times 1.5 = 6$（貫/匹）

總錢：$500 \times 6 = 3000$（貫）$= 3{,}000{,}000$（文）

茶斤數：$\dfrac{3000000}{300} = 10000$（斤）

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# Notes

## Units of Measurement

The problems in these fascicles employ traditional Chinese units:

| Unit | Chinese | Approximate Modern Value |
|---|---|---|
| Length (丈) | zhang | ≈ 3.33 metres |
| Length (尺) | chi | ≈ 33.3 centimetres |
| Length (寸) | cun | ≈ 3.33 centimetres |
| Area (平方丈) | square zhang | ≈ 11.09 square metres |
| Volume (立方丈) | cubic zhang | ≈ 37.0 cubic metres |
| Volume (立方尺) | cubic chi | ≈ 37.0 litres |
| Capacity (石) | shi | ≈ 60 kilograms (grain) |
| Capacity (升) | sheng | ≈ 0.6 litres |
| Weight (斤) | jin | ≈ 600 grams (Song dynasty) |
| Weight (兩) | liang | ≈ 37.5 grams |
| Weight (錢) | qian | ≈ 3.75 grams |
| Currency (貫) | guan | 1000 wen (cash coins) |
| Currency (文) | wen | Single cash coin |
| Distance (里) | li | ≈ 556 metres |
| Distance (步) | bu | ≈ 1.54 metres |

## Mathematical Methods

The *Shùxué Jiǔzhāng* employs several characteristic mathematical techniques:

1. **Frustum volume formula:** For a rectangular frustum with upper dimensions $a \times b$, lower dimensions $c \times d$, and height $h$:
$$V = \frac{h}{3}(ab + cd + ad + bc)$$
This is equivalent to the modern prismatoid formula.

2. **Proportional distribution (衰分):** Problems involving allocation according to given ratios.

3. **Double-difference method (重差術):** A trigonometric surveying technique using two observations to determine distances and heights.

4. **Conversion between currencies:** The “bundle” (陌) system of currency conversion, reflecting the debased coinage of the Southern Song period.

## Historical Context

Qin Jiushao completed the *Shùxué Jiǔzhāng* in 1247, during the turbulent final decades of the Southern Song Dynasty. The military problems in Fascicle 8 directly reflect the desperate military situation of the time, as the Song faced existential threats from the Mongols under Genghis Khan and later Kublai Khan.

The trade problems in Fascicle 9 reflect the sophisticated commercial economy of the Song, with its paper currency, government monopolies on salt and tea, and extensive long-distance trade networks.

*— End of Fascicles 7–9 —*